Wardat Shams Iqbal

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EDUCATION

BRAC University – B.Sc., Computer Science **Sep 2022 – Aug 2026**
Cumulative GPA: **3.96 / 4.00**.

Undergraduate Thesis. *Rethinking Tokenization in Large Language Models: Efficacy, Byte-Level Alternatives, and Detecting Under-trained Tokens.* Supervised by Dr. Swakkhar Shatabda and Dr. Farig Yousuf Sadeque.

- Co-authored a controlled Qwen2.5-7B repair study using Qwen2.5-14B as a reference. Output-row norm rescaling repaired 604 of 711 baseline-broken tokens while increasing rare-token intrusion from 0.0010 to 0.0644.

PROJECTS

MindWell — Employee Wellness Platform (team project). *FastAPI, PostgreSQL, SQLAlchemy, Alembic.*
- Implemented the questionnaire and escalation modules across FastAPI routes, SQLAlchemy models, Alembic migrations, service logic, schemas, and a React/TypeScript questionnaire flow.
- Wrote five branching rules over PHQ-9-style questions so session length adapts between 5 and 10 questions based on earlier answer scores.

Text2SQL Analytics. *Python, FastAPI, PostgreSQL, Docker Compose, Gemini API.* (Repo)
- Built a FastAPI service that turns natural-language questions into read-only Postgres queries through Gemini, covering the whole path from prompt construction to safe execution.
- Wrote a statement-level guard layer that rejects DML and DDL before any query reaches the database, with per-query row limits and statement timeouts on a read-only database role. Query restrictions are covered by pytest.

Local GPT-2 (355M) Pre-training Engine. *Python, PyTorch, CUDA.* (Repo)
- Ran a single-GPU PyTorch pre-training pipeline for randomly initialized GPT-2 Medium (355M).

TCGA PANCAN Gene Expression Classifier. *Python, scikit-learn, Optuna.* (Repo)
- Built a classification pipeline that separates five cancer types from high-dimensional RNA-Seq gene expression data.
- Wrote a sequential feature-reduction workflow (variance thresholding, ANOVA selection, then PCA) that compresses over 20,000 features into 200 components.

EXPERIENCE

Researcher, MIRAI Lab, University of Dhaka **Starting Nov 2026**
- Selected to join the lab under Dr. Md Mamun-or-Rashid for assigned NLP, computer vision, and related research and development projects, with responsibilities covering project deliverables, research meetings, and documentation.

Undergraduate Teaching Assistant, BRAC University CSE **Feb 2025 – Jan 2026; Jul – Sep 2026**
- Tutored over 200 undergraduate students in Data Structures, Algorithms, and Object-Oriented Programming across four semester appointments, with roughly 15 hours a week of consultation and code review.
- Reviewed CSE220 lab assignments and prepared sampled submissions and records for later accreditation assessment during the first appointment.

TECHNICAL SKILLS

Languages: Python, Java, C / C++, SQL.
Backend & APIs: FastAPI, Flask, Express.js, REST API design, SQLAlchemy, Alembic migrations.
Databases: PostgreSQL, MySQL, MongoDB.
Machine Learning & NLP: PyTorch, scikit-learn, LLM pre-training and fine-tuning, BERT fine-tuning, custom loss design, multi-task learning, BPE tokenization, multilingual and low-resource modelling.
Tooling: Docker, Docker Compose, Git, GitHub, pytest.

PUBLICATIONS

Wardat Shams Iqbal, Ruwad Naswan, Swakkhar Shatabda. *CLRG at SemEval-2026 Task 3: One Size Does Not Fit All: A Resource Adaptive Framework for Dimensional Sentiment Regression.* Proceedings of SemEval-2026, co-located with **ACL 2026**. **First author.** aclanthology.org/2026.semeval-1.404